

Digital Signal Analysis and Processing Exam Notes

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

4. Discrete Filter Structures

8 hrs · 68 questions across the 45 papers · worth about 10 of 80 marks · 449 marks, 12.3% of the archive · 407 numerical (91%) + 42 theory · 3 syllabus sub-topics · 5th of the 8 chapters by weight, and tied with Ch 1 as the longest at 8 hours.

Read this first:

- 42 of the 68 questions are the **lattice** TOP 42/45, worth 314 of the chapter's 449 marks. It is one calculation, the **step-down recursion**, run on whatever polynomial the paper prints. Learn it once and two thirds of the chapter is finished.
- The other two topics are **realization structures** TOP 16/45 (direct form I and II, cascade, parallel) and **quantization** TOP 10/45 (number formats, round-off, limit cycles).
- **Nothing in this chapter changes what the filter does.** Every structure here realizes the *same* $H(z)$. They differ in how much memory they need, how many multiplies, and how badly finite word length hurts them. That sentence is the answer to "why do we need different structures".
- **The lattice is stable exactly when every $|k_m| < 1$.** That single line answers the "check stability" half of a dozen questions, and it needs no root finding at all.
- A trap worth knowing before you start: a **symmetric** polynomial gives $|k_N| = 1$, and the recursion divides by $1 - k_N^2$. **A linear-phase FIR filter has no lattice realization.** Five papers ask for one anyway; see §4.4.

4.1 Why there is more than one structure

Why do we need different realization structures for the same system? 2 4

the opening paragraph for any structure question

What is the same

- Every structure in this chapter realizes the **same** $H(z)$, so it has the same impulse response, the same poles and zeros, and the same frequency response. With infinite precision they are indistinguishable.
- So a question that asks you to "realize" a system is asking for a **block diagram or signal flow graph**, not for new mathematics.

What differs, and why it matters

- **Memory:** how many delay elements. Direct Form II uses the minimum possible, $\max(M, N)$.
- **Arithmetic:** how many multiplies and adds per output sample.
- **Finite word length:** how far the poles move when the coefficients are rounded to b bits, and how much round-off noise accumulates. **This is the real reason** cascade and parallel forms are preferred in practice: a direct form of high order is extremely sensitive, because every pole depends on every coefficient.
- **Modularity:** cascade and lattice are built from identical sections.

The three building blocks, and the two ways to draw them

- Every structure is made of **unit delays** z^{-1} , **multipliers** (a constant on an arrow) and **adders**.
- A **block diagram** draws adders as circles with $+$; a **signal flow graph** draws nodes and directed branches, with the gain written on the branch. They carry the same information, and papers ask for either.
- **Transposition:** reverse every arrow, swap adders with branch nodes, and exchange input with output. The transposed structure has the *same* $H(z)$. **No paper in the 45 asks for a transposed form**, and no source in the subject covers it, but it is worth one line because it explains how two diagrams that look nothing alike can be the same filter.

4.2 Direct Form I and Direct Form II

Obtain the Direct Form I and Direct Form II realization of the given system 2+2 4 5

TOP 16/45 79 Bh 2+2, 79 Ba 4, 82 Bh 2+2, 74 Ch 5, 72 Ch 5, 72 Ka 4, 69 Ch 4, 68 Bh 9, 70 Ch 4 · worked in §1 of the numerical companion

Direct Form I: the difference equation, drawn

Start from

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k]$$

- Draw the **feedforward** part first: a chain of M delays down the input, each tapped by b_k into a summing node.
- Then the **feedback** part: a chain of N delays down the output, each tapped by $-a_k$ into the same node.
- **It is the equation copied onto paper**, which is why it is the easy 2 marks.
- Cost: $M + N$ delays, $M + N + 1$ multiplies.
- Read it as $H(z) = B(z) \cdot \frac{1}{A(z)}$: all-zero section first, all-pole section second.

Direct Form II: swap the halves and share the delays

Because the two sections are LTI, they commute:

$$H(z) = \frac{1}{A(z)} \cdot B(z)$$

- Put the **all-pole section first**. Its output is an intermediate signal $w[n]$:

$$w[n] = x[n] - \sum_{k=1}^N a_k w[n-k], \quad y[n] = \sum_{k=0}^M b_k w[n-k]$$

- Now **both** chains delay the same signal $w[n]$, so **one chain of delays serves both**.
- Cost: $\max(M, N)$ delays, which is the fewest any structure can use. A structure with the minimum number of delays is called **canonic**.
- **This is the only reason Direct Form II exists**: same arithmetic, half the memory.

Drawing it, in four steps

1. Write the equation with $y[n]$ alone on the left and **watch the signs**: every a_k crosses over and becomes $-a_k$.
2. Count N (highest y delay) and M (highest x delay). The chain has $\max(M, N)$ delays.
3. Down the left of the chain put $-a_1, -a_2, \dots$ feeding back into the top adder. Down the right put $b_0, b_1, \dots$ feeding forward into the output adder.
4. Label $w[n]$ at the top of the chain. A missing coefficient is a **missing branch**, not a zero multiplier: if the paper's equation skips $x[n-1]$, leave that tap out.

If $a_0 \neq 1$, **divide the whole equation by a_0 first**. 74 Ch prints $3y[n] + y[n-1] + 2y[n-4] = 2x[n] + x[n-3]$, and every coefficient must be divided by 3 before drawing. Forgetting this is the most common lost mark in §4.2.

4.3 Cascade and parallel forms

Realize the given $H(z)$ in cascade form of second-order sections, and in parallel form 4 5 6 7

TOP 16/45 81 Ba 5, 74 Ma 7, 80 Ba 4, 76 Ch 4, 77 Ch 7, 69 Bh 6, 67 Mng, 78 Ch 6

Cascade: multiply the sections

$$H(z) = b_0 \prod_{k=1}^K \frac{1 + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}}$$

- Group the poles and zeros into **second-order sections**, each realized in Direct Form II, and chain them output to input.
- **A complex pole must be paired with its conjugate** in the same section. That is the whole point: the section then has **real** coefficients, $a_{1k} = -2\text{Re}(p)$ and $a_{2k} = |p|^2$.

Parallel: add the sections

$$H(z) = C + \sum_{k=1}^K \frac{\gamma_{0k} + \gamma_{1k}z^{-1}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}}$$

- Get here by **partial fractions**, exactly as in Chapter 2, then recombine each conjugate pair into one real second-order section.
- C is the constant from the division when $H(z)$ is improper; it is a direct path from input to output.

- Real poles pair with real poles, or sit alone as a first-order section.
- An odd number of real poles leaves one first-order section; that is normal, not an error.
- **Cascade needs factors, parallel needs partial fractions.** That is the only difference in the work.
- Parallel has the best round-off behaviour of the three: an error in one section cannot propagate into the others, because they do not feed each other.

Turning a conjugate pair into real coefficients

- A pair of poles at $re^{\pm j\theta}$ multiplies out to

$$(1 - re^{j\theta}z^{-1})(1 - re^{-j\theta}z^{-1}) = 1 - 2r \cos \theta z^{-1} + r^2 z^{-2}$$

- So a section written $1 - 2r \cos \theta z^{-1} + r^2 z^{-2}$ needs only r and θ , both of which the papers usually hand you directly.
- For a pair given in rectangular form $\sigma \pm j\omega$, it is $1 - 2\sigma z^{-1} + (\sigma^2 + \omega^2)z^{-2}$.
- **Check every section has real coefficients before drawing.** A surviving j means a pole was left unpaired.

4.4 The FIR lattice

Compute the lattice coefficients and draw the structure for the given $H(z)$; check stability **5 6 8 10**

TOP 42/45 80 Bh **10**, 81 Ch, 72 Ash **4+2+1**, 79 Ch, 69 Bh **8**, 71 Bh **6+2**, 70 Ma **6**, 71 Ch **4+2+1**, 81 Ba **5**, 78 Bh **3+7**, 79 Bh **6**, 71 Shr **5**, 80 Ch, 74 Ma **7**, and 25 more · **the single most valuable calculation in the subject**

What the structure is

An M -stage lattice chains identical two-input two-output sections. Stage m takes a forward signal f_{m-1} and a backward signal g_{m-1} and produces

$$f_m[n] = f_{m-1}[n] + k_m g_{m-1}[n-1]$$

$$g_m[n] = k_m f_{m-1}[n] + g_{m-1}[n-1]$$

with $f_0[n] = g_0[n] = x[n]$ and $y[n] = f_M[n]$.

- The k_m are the **reflection coefficients**, one per stage.
- The forward path realizes $A_M(z)$; **the backward path realizes its reverse** $z^{-M}A_M(z^{-1})$, which is what makes the all-pass of §4.5 so tidy.
- Only **one** multiplier value per stage, used twice. That symmetry is why the lattice is far less sensitive to coefficient quantization than a direct form, and why it is used for speech.

The step-down recursion (**direct form** → **lattice**)

Given $A_M(z) = 1 + a_M(1)z^{-1} + \dots + a_M(M)z^{-M}$:

$$k_m = a_m(m)$$

$$a_{m-1}(i) = \frac{a_m(i) - k_m a_m(m-i)}{1 - k_m^2}, \quad i = 1, \dots, m-1$$

1. Read off k_M : it is the last coefficient, free.
2. Fold A_M against its own reverse, divide by $1 - k_M^2$, and you have A_{M-1} , one degree lower.
3. Repeat down to A_1 , whose single coefficient is k_1 .

The step-up recursion (**lattice** → **direct form**)

$$a_m(i) = a_{m-1}(i) + k_m a_{m-1}(m-i), \quad a_m(m) = k_m$$

Same relation, rearranged. Papers ask in both directions: 79 Bh and 71 Shr give $k = \{\frac{1}{4}, \frac{1}{2}, \frac{1}{3}\}$ and want the FIR coefficients.

Stability, in one line

- **The system is stable if and only if $|k_m| < 1$ for every m .**
- No root finding, no Jury table. The step-down is the Schur-Cohn test: it is testing whether every root of $A(z)$ lies inside the unit circle.
- A question that says "check stability" after asking for the lattice is asking you to **look at the numbers you already computed**. It is a free mark.

The symmetric trap: why five papers ask for an impossible lattice If $A(z)$ is **symmetric**, meaning its coefficient list reads the same backwards, then $a_M(M) = a_M(0) = 1$, so $k_M = 1$ and the step-down divides by $1 - k_M^2 = 0$. **A symmetric polynomial has no lattice realization.** And symmetry is exactly the linear-phase condition $h[n] = h[M-n]$ from Chapter 3, so a **linear-phase FIR filter can never be put in lattice form**. The papers that hit this are **72 Ch** ($1 + 2z^{-1} + z^{-2}$), **73 Bh** ($1 + 2z^{-1} + 2z^{-2} + z^{-3}$), **75 Ch** and **73 Ch** (both with $1 + \frac{2}{3}z^{-1} + \frac{5}{8}z^{-2} + \frac{2}{3}z^{-3} + z^{-4}$), and **74 Ash**, whose last coefficient is -1 . Say so, show the division by zero, and note that the zeros sit on the unit circle. It is a better answer than a page of arithmetic that cannot finish.

4.5 Lattice-ladder, for a pole-zero system

Why a ladder is needed

- The plain lattice realizes an **all-zero** $A_M(z)$, or upside down an **all-pole** $1/A_N(z)$.
- A general $H(z) = B(z)/A(z)$ needs one more piece: tap every backward signal $g_m[n]$ through a weight C_m and sum them.

$$y[n] = \sum_{m=0}^M C_m g_m[n]$$

- The lattice supplies the poles, **the ladder taps supply the zeros**.

Getting the ladder coefficients

Run the step-down on the **denominator** $A(z)$ and keep every intermediate polynomial $A_i(z)$. Then, working **downwards** from $m = M$:

$$C_m = b_m - \sum_{i=m+1}^M C_i a_i(i-m)$$

1. $C_M = b_M$, because the sum is empty.
2. Each lower C subtracts what the higher taps already contribute.
3. **The k 's come only from $A(z)$** , so two papers with the same denominator have the same lattice and differ only in the ladder. 76 Ch and 76 Ash are such a pair, and so are 74 Bh and 75 Bh.

The all-pass result, and why it is worth remembering 70 Bh gives $H(z) = \frac{1 + 0.4z^{-1} - 1.2z^{-2} + 2z^{-3}}{2 - 1.2z^{-1} + 0.4z^{-2} + z^{-3}}$ and asks how a lattice implements it. Look at the coefficients: **the numerator is the denominator reversed**. That is the definition of an all-pass, $H(z) = z^{-N} A(z^{-1})/A(z)$, and the backward path of a lattice already computes exactly $z^{-N} A(z^{-1})$. So the ladder collapses: every C_m is zero except $C_N = 1$. **An all-pass filter is just a lattice with the output taken from the last backward node**, no ladder taps at all. The reflection coefficients are $k = \{-0.56, \frac{2}{3}, \frac{1}{2}\}$, all less than 1 in magnitude, so it is stable.

4.6 Number representation and quantization

Write about the sign magnitude and 2's complement representation of a binary fractional number. Write about truncation error and rounding error 6 3+4

TOP 10/45 69 Bh 6, 69 Bh 3+4, 75 Ash 3, 67 Mng, 71 Shr 1+1

The three fixed-point codes

A fraction in $[-1, 1)$ with b bits after the binary point. Positive numbers look the same in all three; only negatives differ.

Code	How a negative is formed
Sign magnitude	sign bit 1, magnitude bits unchanged
1's complement	invert every magnitude bit
2's complement	invert every bit, add 1 in the last place

- Worked, for $\pm 5/8$ in 3 bits, $5/8 = 0.101_2$:
 $+5/8 = 0.101$ in all three.
 $-5/8 = 1.101$ (SM), 1.010 (1's), 1.011 (2's).
- **2's complement is used in practice** because addition needs no special case and overflows in a partial sum cancel out.

Rounding against truncation

With step $q = 2^{-b}$ and error $e = Q(x) - x$:

Scheme	Error range
Rounding, any code	$-q/2 \leq e \leq q/2$
Truncation, sign mag.	$-(q - q_u) \leq e \leq (q - q_u)$
Truncation, 2's comp.	$-(q - q_u) \leq e \leq 0$

- **Rounding is better**, and this table is why: its error is centred on zero, so it has **no bias**, and its range is half as wide.
- Truncating a **sign magnitude** number always shrinks the magnitude, so the error is symmetric about zero but takes the sign of x .
- Truncating a **2's complement** number always moves the value **down**, never up, so the error is **one-sided and negative**: a systematic bias, a DC offset that accumulates through a filter.
- $q_u = 2^{-b_u}$ is the step before quantization, with $b_u > b$.

The three places quantization bites

- **Input (A/D) quantization**: modelled as additive white noise of variance $q^2/12$ for rounding.

- **Coefficient quantization:** the a_k and b_k are stored in b bits, so **the poles move**. A high-order direct form is the worst case, because every pole depends on every coefficient; cascade and parallel forms confine the damage to one section, which is the practical argument for §4.3.
- **Product round-off:** each multiply doubles the word length and must be shortened again. In a recursive filter this feeds back, which is §4.7.

4.7 Limit cycles and the dead band

What is the limit cycle effect in a recursive system? Describe it with an example. Define dead band

3 5 1+3

TOP 10/45 71 Ch 3, 70 Asa 1+3, 72 Ma 5, 81 Ch, 80 Ch, 70 Bh 3+4, 66 Ma 4, 71 Shr 1+1+2+1

What a limit cycle is

- In a **recursive** filter the rounded product is fed back and rounded again. Once the signal is small enough, rounding can return the *same magnitude* it started from.
- The filter then never decays: after the input stops it sits in a **self-sustaining oscillation** of fixed amplitude. That is a **zero-input limit cycle**.
- It happens only with quantization. **The ideal infinite-precision filter is perfectly stable**; the limit cycle is created by the arithmetic, not by the poles.
- FIR filters cannot have them: no feedback, nothing to sustain.

The dead band

For the first-order $y[n] = Q\{a y[n-1]\} + x[n]$ with rounding, the filter cannot escape once

$$|y[n]| \leq \frac{q}{2(1-|a|)}$$

- That interval is the **dead band**: inside it, quantization exactly undoes the decay that a should have produced.
- The oscillation amplitude is bounded by the dead band, so a longer word (smaller q) makes it smaller but never removes it.
- If $a > 0$ the trapped value holds **constant**, period 1. If $a < 0$ it **alternates in sign**, period 2.

The worked example, 66 Ma (and the shape of every limit-cycle answer)

$y[n] = Q\{-0.5 y[n-1]\} + x[n]$, rounded to 3 bits so $q = \frac{1}{8}$, with $y[-1] = 0$ and $x[n] = 0.875 \delta[n]$:

n	$-0.5 y[n-1]$	rounded to $\frac{1}{8}$	$+ x[n]$	
0	0	0	0.875	the impulse arrives
1	-0.4375	-0.5	-0.5	
2	0.25	0.25	0.25	
3	-0.125	-0.125	-0.125	now inside the dead band
4	0.0625	0.125	0.125	rounding put it back up
5	-0.0625	-0.125	-0.125	

- From $n = 3$ on it alternates $\pm \frac{1}{8}$ for ever: a limit cycle of amplitude $\frac{1}{8}$ and **period 2**, the period coming from $a < 0$.
- Dead band: $\frac{q}{2(1-|a|)} = \frac{1/8}{2(1-0.5)} = \frac{1}{8}$, which is exactly the amplitude observed. **Quote the formula and the table agrees with it.**
- Step 4 is the whole mechanism in one line: $-0.5 \times (-0.125) = 0.0625$, which is *half* a step, and rounding carries it up to a full 0.125 instead of down to 0.

4.8 Last-minute recall

Say these without thinking

- Direct Form II is **canonic**: $\max(M, N)$ delays, the fewest possible.
- Divide through by a_0 before drawing anything.
- Cascade needs **factors**; parallel needs **partial fractions**.
- A complex pole always sits with its conjugate, giving $1 - 2r \cos \theta z^{-1} + r^2 z^{-2}$.
- Step-down: $k_m = a_m(m)$, then
$$a_{m-1}(i) = \frac{a_m(i) - k_m a_m(m-i)}{1 - k_m^2}.$$
- Step-up: $a_m(i) = a_{m-1}(i) + k_m a_{m-1}(m-i)$.
- **Stable** $\Leftrightarrow$ every $|k_m| < 1$.
- Ladder: $C_m = b_m - \sum_{i>m} C_i a_i(i-m)$, downwards from $C_M = b_M$.
- Dead band = $\frac{q}{2(1-|a|)}$; $a < 0$ gives period 2.

Mistakes that cost marks

- **Forgetting to divide by a_0** when the paper prints $3y[n] + \dots$.
- Sign errors moving a_k across the equals sign: the feedback taps are $-a_k$.
- Running the step-down on the **numerator** of a lattice-ladder question. The k 's come from $A(z)$, always.
- Grinding through a step-down that has already produced $|k| = 1$. Stop and say the polynomial is symmetric.
- Reporting k to two decimals and then calling $|k| = 0.99$ stable without saying so. Keep fractions where the paper gives fractions.
- Claiming an FIR filter can have a limit cycle. It cannot: no feedback.
- Saying truncation error is symmetric for 2's complement. **It is one-sided**, and that is the point of the question.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads "leads to / therefore", $+$ reads "and", $=$ reads "is". $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months:** 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.